

CONSTANT SPEED STORY: DUAL-GRAPH EDITION

Name: _____ Period: _____ Date: _____

UNIT 2: KINEMATICS • AUTHOR, ILLUSTRATE, SOLVE & GRAPH • DAY 15
PERFORMANCE TASK

SCORE: _____ / 10 pts

Mission: Invent an original character/vehicle in constant speed motion. Illustrate the scene, write your word problem, and solve using the GUESS method. On the back, plot both graphs to prove your solution!

HS-PS2-1 •
PERFORMANCE
TASK

PART 1: SCENARIO ILLUSTRATION

Sketch your moving object with origin (x = 0) and motion arrow (→)

[Draw your character, vehicle, runner, robot, animal, or space probe in steady motion]
Label: Starting Reference Point (x = 0 m) • Direction of Travel (→) • Estimated Speed (v)

PART 2: ORIGINAL KINEMATICS WORD PROBLEM

Give 2 known values with units, and ask for the 3rd unknown variable

Prompt: Write a clear word problem describing your scene. Choose which unknown to solve for: **Speed (v)**, **Distance (d)**, or **Time (t)**.

PART 3: MATHEMATICAL SOLUTION PROOF (G.U.E.S.S. METHOD)

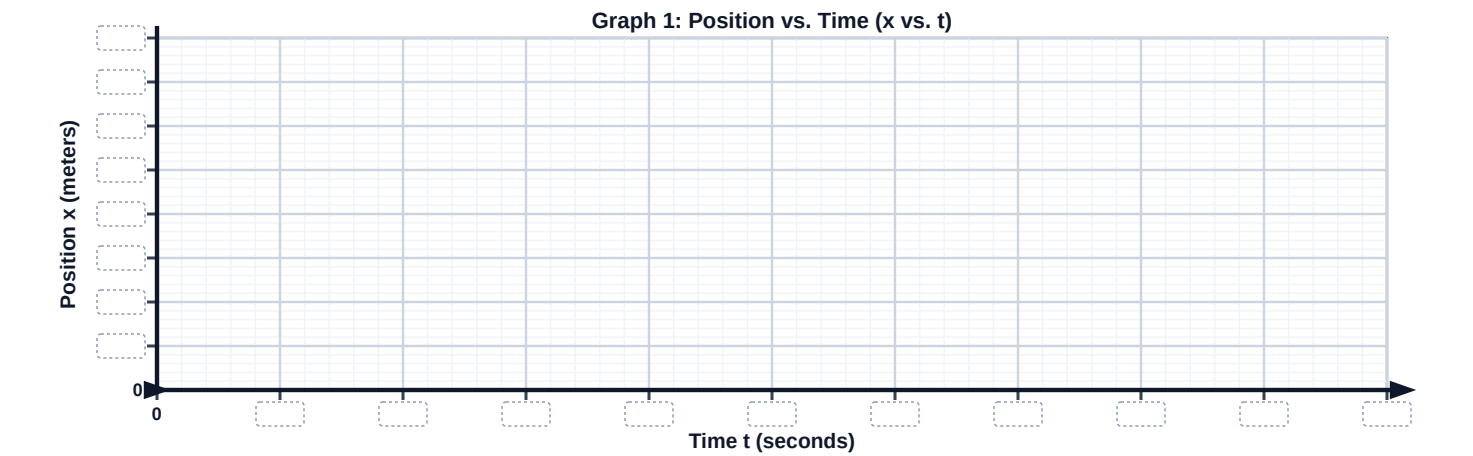
Show complete algebraic steps with explicit units

G	Givens (with units)	Known #1: _____	Known #2: _____	Unit Conversions (if any): _____
U	Target Unknown	Variable to solve for: ☐ Speed (v) ☐ Distance (d) ☐ Time (t)		
E	Equation (Formula)	Circle formula used: v = d / t d = v • t t = d / v		
S	Substitute	Plug numbers and units into your formula: _____		
S	Solve & Box	Final numerical answer with units:		

Horizontal Scale (Time t):
Total Time = _____ s • Each grid block = _____ s

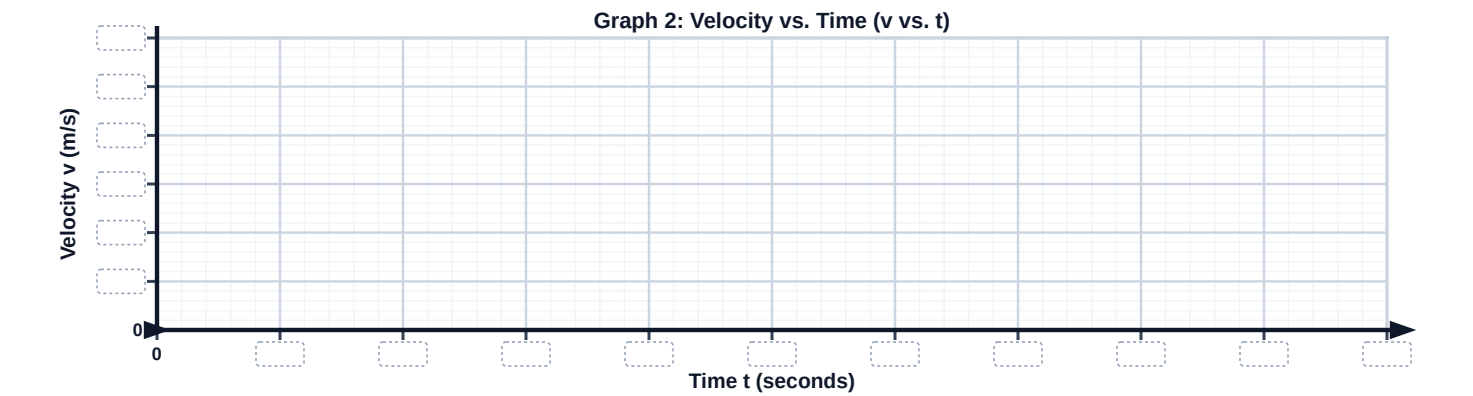
Graph 1 Vertical (Position x):
Total Dist = _____ m • Each grid block = _____ m

Graph 2 Vertical (Velocity v):
Speed v = _____ m/s • Each grid block = _____ m/s



Graph 1 Slope Proof: Slope = $\Delta x / \Delta t = (\text{_____ m} - 0 \text{ m}) / (\text{_____ s} - 0 \text{ s}) = \text{_____ m/s}$

Matches my story speed? ☐ YES ☐ NO



Graph 2 Area Proof: Shaded Area = Height $\times$ Width = (_____ m/s) $\times$ (_____ s) = _____ meters

Matches my story distance? ☐ YES ☐ NO

DUAL-GRAPH SYNTHESIS • CONNECTING SLOPE AND AREA

CORE CONCEPT CHECK

1. Slope Physical Meaning:
The visual steepness (slope) of your Position-Time line physically represents the object's _____.

2. Area Physical Meaning:
The shaded geometric area under the Velocity-Time line physically represents the total _____ traveled.

3. What If Challenge: If your object traveled at *twice* its speed in the same time, describe what would happen to: (a) Graph 1's slope: _____ • (b) Graph 2's shaded area: _____